

Demand Forecasting SaaS — Final Release Summary

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Comprehensive summary of all Sprints (0 → 13) — architecture, deliverables, compliance, and QA readiness.

Project Overview

- The Demand Forecasting SaaS MVP was developed through 14 structured sprints (0–13) covering planning, backend, frontend, and deployment.
- The platform enables dataset upload, automatic conversion, forecast visualization, and insights generation, supporting multiple data sources and models.
- Each sprint delivered testable and independently deployable components, culminating in a full enterprise-ready system with comprehensive documentation.

Sprint 0 – Foundation

- Project setup: repo, folder structure, ESLint/Prettier/Tailwind/TypeScript.
- Backend initialized with FastAPI + PostgreSQL (Neon).
- Docker Compose dev environment created.

Sprint 1 – Core Backend

- Implemented FastAPI endpoints, database connection pool, and health checks.
- Set up RLS policies and PostgreSQL schema migrations.

Sprint 2 – Dataset Upload & Presign

- Implemented presign endpoints for file uploads.
- Integrated Cloudflare R2 / MinIO adapters for storage.

Sprint 3 – Conversion Worker

- Added worker to auto-convert Excel/Google Sheets to CSV.
- Enforced free-tier retention (delete originals).

Sprint 4 – Forecast Chart & Insights

- Implemented chart rendering and insights API.
- Used LLM provider for contextual forecast analysis.

Sprint 5 – LLM Provider Management

- Added llm_providers tables, plan mappings, and admin override UI.
- Supported Gemini for Free/Pro and OpenAI for Business/Enterprise tiers.

Sprint 6 – Multi-region DB & GDPR RLS

- Neon Postgres setup with regional instances and RLS policies.
- Implemented JWT/JWKS middleware with session-local org_id/user_id.

Sprint 7 – Quotas & Enforcement

- Added rate limiting, quota checks per plan, and response codes 403/429.
- Implemented Redis-based counters.

Sprint 8 – Retention & Purge Jobs

- Added cron job to purge old datasets.
- Implemented safe-delete and soft-delete retention model.

Sprint 9 – Webhook Security & Retry

- Introduced HMAC verification for webhooks.
- Implemented retry logic with exponential backoff.

Sprint 10 – Terraform & Infra Templates

- Provided IaC for Hetzner servers, Cloudflare R2, Neon DB, and Supabase Auth.
- Added .env.example and CI/CD setup scripts.

Sprint 11 – Monitoring & Alerts

- Configured Prometheus/Grafana dashboards and Sentry integration.
- Added alert rules for job failures, quota breaches, and LLM token spikes.

Sprint 12 – Trust & Transparency UI

- Added Trust Center, Privacy Policy, Audit Logs, Residency and Retention Views.
- Implemented Delete Org Workflow with GDPR-compliant deletion.

Sprint 13 – Release Finalization

- Added Makefile, dev_setup.sh, verify scripts, and admin seeding tools.
- Included GitHub Actions workflows (self-hosted and hosted runners).
- Generated comprehensive docs, PDFs, and deployment scripts.

Compliance & QA Readiness

- GDPR-compliant: data deletion, user consent, region awareness.
- SOC2 alignment: audit logging, access control, monitoring.
- Multi-tenant isolation: PostgreSQL RLS + SET LOCAL org_id per session.
- Encryption: TLS enforced at API and database layers.
- Sentry logging and Prometheus alerting integrated.
- Manual QA pending (to be done post full sprint delivery).